Biocybernetics and Biomedical Engineering 44 (2024) 218–230

Contents lists available at ScienceDirect

Biocybernetics and Biomedical Engineering

journal homepage: www.elsevier.com/locate/bbe

Research Paper

A comprehensive analysis of task-specific hand kinematic, muscle and force synergies

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ARTICLE INFO

Keywords:
Force synergies
Kinematic synergies
Muscle synergies
Hand
Biomechanics
Prosthetics

ABSTRACT

Synergies were demonstrated to exist in the kinematic, force and muscular domains, and their task-specificity and subject-specificity was also highlighted in literature. Despite that, no works have extracted synergies on specific grasp classes to analyze task-specific synergistic patterns. Moreover, only few studies focused on the combined analysis of kinematic, force and muscle synergies.

The aim of this work was to (i) identify the grasp classes on which to extract task-specific synergies; (ii) extract subject-specific and task-specific synergies in the three domains and (iii) calculate the similarity of the extracted synergies among subjects and define average generalized synergies.

8 subjects were recruited to perform 21 reach-to-grasp tasks and the kinematics, contact forces and muscular activation of the hand were acquired. A LDA classifier allowed distinguishing power and precision grasp classes with an average accuracy of 89% considering kinematic data alone and combined kinematic, muscle and force data. Subject and task-specific synergies were therefore extracted on these two classes. Kinematic and force synergies were distinctive for the two classes, and highly similar among subjects, thus suggesting the possibility of adopting generalized synergies to describe grasp strategies. Conversely, muscle synergies did not differ particularly for the two classes. The combined analysis of force and kinematic data suggested that the hand posture may be somehow modulated by the optimal distribution of contact forces to perform stable grasps. Simulations with a virtual hand confirmed that stability significantly increased when grasps were generated by activating combined kinematic and force synergies rather than kinematic synergies only.

1. Introduction

The hand is the most sophisticated and versatile district of the body, capable of performing infinite types of movements. Given the critical role the hand has in Activities of Daily Living (ADLs), understanding how the Central Nervous System is able to control all 22 degrees of freedom (DoFs) of the hand (23 if the wrist is also considered), is an essential task as well as an engaging challenge.

The concept of performing a particular motor task in different ways is referred to as the redundancy problem and was first described by Bernstein in 1967 [1]. Even though redundancy offers different possibilities to perform any motor act, it complicates the task of the Central Nervous System by forcing it to choose among several potentially competing motor strategies. One of the possible solutions to overcome this problem is to reduce the number of total DoFs to control with respect to those required to perform the task efficiently. This is the basis for the concept of synergies, that refer to a set of relatively independent DoFs that behave as a single functional unit. Synergies can be extracted by applying different dimensionality reduction techniques such as Principal Component Analysis (PCA), non-negative matrix factorization (NNMF), independent component analysis, factor analysis, autoencoders, etc. [2–5]. Among these methods, linear dimensionality reduction techniques were mostly exploited to extract synergies in different domains due to their limited computational burden and ease of interpretation. In particular, PCA is a linear technique that has been widely exploited in literature to extract kinematic [6–12], force [13–15] and muscle [16–18] synergies. It allows reducing the dimensionality of a dataset composed of correlated variables by computing linear combinations of the original variables, i.e. the Principal Components (PCs), also referred to as synergies in the literature. These new uncorrelated variables generate a novel coordinate system (i.e. the

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https://doi.org/10.1016/j.bbe.2024.01.006
Received 31 July 2023; Received in revised form 11 November 2023; Accepted 31 January 2024
Available online 15 February 2024
0208-5216/© 2024 The Author(s). Published by Elsevier B.V. on behalf of Nalecz Institute of Biocybernetics and Biomedical Engineering of the Polish Academy of Sciences. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

synergy space) in which to express data, and are ordered in such a way that the first few synergies retain most of the variation present in the dataset.

Kinematic (or postural) synergies can be defined as linear combinations of joint angle configurations that regulate the shape of the hand in grasping tasks, and were first investigated by Santello et al. [6] in 1998, who demonstrated that the control of the hand to execute grasping tasks involved the combination of three postural synergies that were able to describe around 90% of the original data variance. This study was followed by many others that underlined the task-specificity and subject-specificity of synergies [8,10,12,19,20]. These works suggested that different subjects could exploit different strategies to grasp the same object, and that those objects required different combinations of synergies to be grasped. An attempt in identifying shared synergies across subjects in grasping tasks was presented in [8,10,11]. These works used hierarchical clustering to prove that some synergies were shared across a wide population of subjects, and some other synergies were subject-specific.

Muscle synergies represent a coordinated recruitment of muscle groups that behave as a single functional unit. As the existence of such muscle synergies is now well established [21], some studies tried to investigate whether a relation between kinematic and muscle synergies existed. Weiss and Flanders [16] provided a description of hand postures associated with the American Sign Language alphabet and grasping tasks of everyday objects with a set of muscle synergies, and concluded that there is no one-to-one relation between activated muscles and muscle synergies or kinematic synergies. Tagliabue et al. [17] investigated the interrelation between kinematic and muscle synergies during two-digit grasping. They discovered that, at least for the specific task considered, kinematic synergies could origin in synergistic muscle activation. Finally, Castellini et al. [18] confirmed the existence of kinematic and muscle synergies, extracted during grasping tasks of five different objects, and observed that different grasps could be distinguished in the synergy space, and this was more evident for kinematic than muscle data. Moreover, they discovered that grasps which required similar muscle activation did not necessarily match grasps which corresponded to similar finger positions.

Finally, force synergies represent characteristic correlation patterns of contact force distribution among the fingers and palm during grasping tasks. In the work by Santello and Soechting [22], the existence of a coordination between the normal contact forces exerted by the fingers was investigated, and the authors discovered two main force synergies in the frequency space. In the study by Naceri et al. [14], the authors pointed out the existence of a relation between the finger position and the normal contact force exerted by the thumb and long fingers during a grasping task of a cuboid object. Starke et al. [15] analyzed force synergies during grasping tasks of 14 different objects by monitoring contact forces at the fingers and palm, and revealed the existence of a correlation between contact forces and contact points. In particular, 8 synergies able to explain more than 90% of the data variance were extracted. Moreover, the authors pointed out the task-dependency of contact force patterns. The force patterns in cylindrical grasps and the concept of error compensation were examined in [13] to quantify the synergistic patterns and coordination of multi-finger movements. Force synergies were confirmed to exist in static and dynamic cylindrical grasps. In addition, the authors discovered that if some fingers did not contact the objects during the grasp, those nearby increased their strength to maintain the same level of grasp performance with minimal disruption.

Despite many studies suggested that synergies are task-specific, to the best of our knowledge no studies have in depth investigated the characteristics of those synergies for specific grasp classes. The extraction of task-specific synergies could allow identifying distinctive kinematic, force and muscle patterns of the hand in executing specific grasping tasks. We hypothesize that grasps belonging to different grasp classes can be associated to distinctive hand postures, force patterns and muscle activation, and that the extraction of task-specific synergies could allow representing with more fidelity the hand kinematics and dynamics associated to a particular task or object to be grasped.

In addition, the study of the existing literature works pointed out that only few papers focused on the combined analysis of kinematic, force and muscle synergies, to highlight possible relations between different variables involved in the grasping action, even though the discovery of such relations could allow understanding the strategies the Central Nervous System exploits to generate optimized grasping tasks. In particular, no works describing the relation between kinematic and force domains were found, except for some works that analyzed in general the kinematics and the force of the hand.

Therefore, the first objective of this work was to select the best grasp subdivision on which to further extract subject-specific and task-specific synergies. To do so, we investigated to what extent synergies extracted in the kinematic, force and muscle domains and in combined domains could be representative of the grasp classes identified by the Cutkosky grasp taxonomy [23], which classified grasps in two major classes, i.e., power and precision, that are in turn divided into prismatic and circular. After reviewing previous studies that extracted kinematic, muscle and force synergies and that compared the performance of different dimensionality reduction methods [2], we chose to apply the Expectation-Maximization (EM) PCA [24], which allows performing PCA on datasets affected by measurement noise or missing values, to retrieve a synergy space optimized to describe the true underlying signal variations. We applied EM-PCA to "pure" and "combined" datasets including trials performed by all the volunteers, and we represented data in the synergy space. Then, we analyzed the accuracy of a linear classifier in distinguishing between different grasp classes starting from data expressed in the synergy space in different domains, and we selected the grasp subdivision on which to extract synergies.

Once the grasp classes on which to perform the analysis were identified, the main objective of this work was to in depth analyze the characteristics of subject-specific postural, force, muscle synergies, and combined synergies, i.e., synergies extracted on combined datasets, for specific grasp classes, and to define generalized synergies shared across volunteers.

This analysis was conducted by first calculating subject-specific and task-specific synergies in all "pure" and "combined" domains and then by comparing subject-specific synergies by means of the Cosine Similarity index, in order to identify synergies shared among subjects by taking into account the individual motion strategy of each participant, that is, by considering the hierarchy of the variance expressed by each synergy. Synergies extracted in "pure" and "combined" domains were then analyzed to look for relations between different variables involved in the grasping action.

The rest of the paper is organized as follows: Section 2 reports the materials and methods, Section 3 illustrates the obtained results, that will be discussed in Section 4. Finally, conclusions are provided in Section 5.

2. Materials and methods

2.1. Experimental setup

The experimental setup used in this work allowed recording synchronously the kinematics of the hand, the force exerted by the hand fingers and palm during the interaction with real objects, and the muscular activation of seven properly selected muscles. The overall experimental setup is shown in Fig. 1.

2.1.1. Hand modeling and grasp selection

The human hand-wrist was modeled as a 23-DoFs kinematic structure in which the abduction–adduction (AA) and flexion-extension (FE)

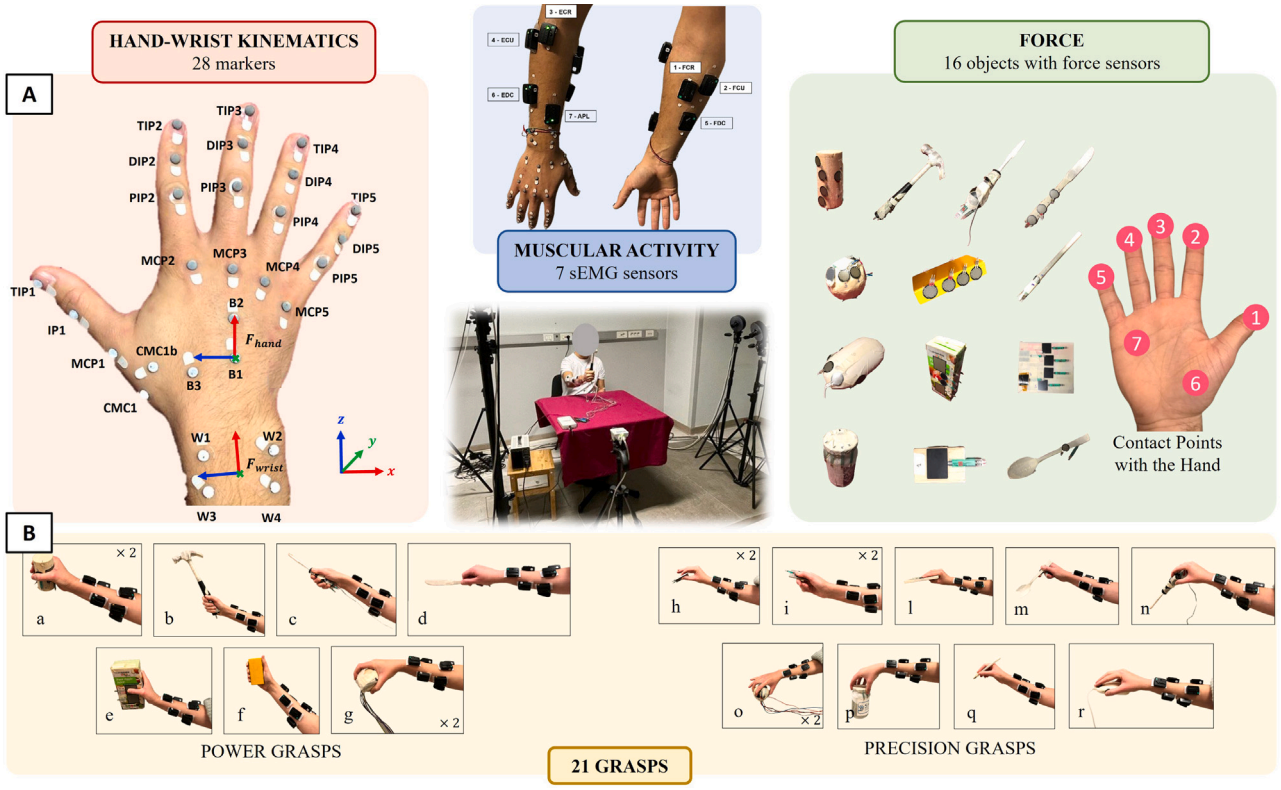


Fig. 1. Experimental Setup. (A) Position of the markers (Hand Kinematics box), of the sEMG sensors (Muscular Activity box) and of the contact points (Force box) on the hand of the subject. In the Force box, the selected 16 objects with force sensors are reported. They were a cylinder, a hammer, a screwdriver, a knife, a sphere, a parallelepiped which simulated a smartphone, a pen, a mouse, a carton of juice, a CD-ROM case, a jar cap, a small squared parallelepiped which simulated a credit card and a key, and a spoon. The cylinder, sphere and small squared parallelepiped were used in both heavy and light versions, but only one version is reported in the figure for brevity. These objects were used to perform 21 grasps shown in B: (a) cylinder, (b) hammer, (c) screwdriver, (d) knife, (e) carton of juice were used to perform power prismatic grasps; (f) parallelepiped (i.e., smartphone) and (g) sphere were used to perform power circular grasps, (h) small squared parallelepiped (i.e., credit card) and (i) small squared parallelepiped (i.e., key) were used to perform precision prismatic pinch and lateral grasps, respectively; (l) CD-ROM case was used to perform a precision prismatic grasp; (m) spoon to perform precision prismatic lateral grasp; (n) screwdriver was used to perform precision circular tridigital grasp; (o) light and heavy spheres were used to perform precision circular tridigital grasps; (p) jar cap was used to perform precision circular pentadigital grasp; (q) pen was used to perform precision circular tridigital grasp and (r) mouse was used to perform precision circular pentadigital grasp.

angles of all fingers were considered. In particular, the kinematic chain of the long fingers was composed of spherical joints with orthogonal and intersecting axes at the metacarpophalangeal (MCP) joints, and of two hinge joints at the proximal and distal interphalangeal (PIP and DIP) joints for each finger. The thumb was modeled as a 5-DoFs kinematic chain with two spherical joints with orthogonal and intersecting axes at the carpometacarpal (CMC) and MCP joint, and a hinge joint at the interphalangeal (IP) joint. In addition, the wrist was modeled as a spherical joint with intersecting and orthogonal axes that allow the FE and the radio-ulnar deviation (RUD) movements.

In order to identify the grasps to be included in the experimental acquisitions, the objects used by Santello et al. in [6], those used in the Hannes hand experimental trials [9] and those present in the Southampton Hand Assessment Procedure (SHAP) [25] were compared, and objects that were common were identified. In addition to the objects found in the mentioned articles, a mouse grasp and a smartphone grasp were also included to consider objects used in the present ADLs. In particular, the 16 selected objects allowed representing all the grasp types included in the Cutkosky grasp taxonomy [23] and executing 21 grasps in total. The 16 objects are shown in Fig. 1A, and the 21 grasps grouped according to the taxonomy defined by Cutkosky are shown in Fig. 1B. In particular, some objects were used to perform more than one type of grasp: spheres were used for both power and tridigital precision grasps; light and heavy squares were used for both pinch grasps and lateral grasps, thus simulating the grasp of a credit card and a key, respectively; the screwdriver was used for both a power and a tridigital precision grasp of an object smaller than the spheres. Moreover, Fig. 1B does not report grasps with both the heavy and light versions of the sphere, cylinder and square, because the grasp to be executed did not differ.

2.1.2. Kinematic analysis

The BTS Smart-D optoelectronic motion capture system was exploited to reconstruct human hand and wrist kinematics. It includes 8 infrared cameras with a resolution up to 1.4 Mp and accuracy ≤ 0.2 mm in a volume of $3 \times 2 \times 2$ m^3. The acquisition frequency was set to 60 Hz. The system allowed recording the positions of 28 retroreflective markers, with a diameter of 6 mm, positioned on specific anatomical landmarks of the hand and forearm (Fig. 1A - Hand Kinematics). In particular, markers on the hand were positioned according to the protocol proposed by Cordella et al. [26]. From these markers, it was possible to define the hand base reference frame F_{hand} and the local frames at each joint. In particular, F_{hand} was defined from markers B1, B2 and B3 as shown in Fig. 1A. The kinematic protocol allowed reconstructing the AA angles of the CMC joint of the thumb, the AA angles of all the MCP joints and the FE angles of the CMC, MCP, IP, PIP and DIP joints with high accuracy [26].

The wrist FE and RUD angles were reconstructed by positioning four markers on the forearm (Fig. 1A) [27]. From these markers, the wrist reference frame F_{wrist} was defined. Wrist angles were extracted as a sequence of Euler angles between F_{hand} and F_{wrist}: the FE was the rotation of the hand reference frame around z_{wrist}, and the RUD angle was the rotation of the hand frame around y_{wrist}.

Table 1
Angles signs.

Angle	+	−
AA Fingers	Toward the thumb	Toward the little
FE Fingers	Flexion	Extension
FE Wrist	Flexion	Extension
RUD Wrist	Radial Deviation	Ulnar Deviation

The monitored angles and the convention of the angles signs are reported in Table 1.

2.1.3. Force analysis

The grasping force during hand-object interaction was monitored by positioning different types of piezoresistive sensors on the 16 selected objects.

In particular, three types of Force Sensing Resistors (FSR) developed by Interlink Electronics, Inc. (Camarillo, California, United States) (i.e., FSR 402, 400 and 406 models) and two types of FlexiForce sensors developed by Tekscan, Inc. (United States) (i.e., A401 and A502) were used. Each sensor was characterized using the Instron testing machine by applying a force in the range 0–8 N and recording the corresponding output voltage. The characterization equation that allowed computing the force was a cubic polynomial for all the sensors.

A variable number of piezoresistive force sensors was positioned on the objects according to the type of grasp to be executed. For the objects grasped with power grasps, 7 sensors were placed: 5 to make contact with the fingers of the hand and 2 to make contact with the palm of the hand. Objects grasped with precision grasps were sensorized with a number of force sensors varying from 2 to 5 depending on whether a bidigital, tridigital or pentadigital grasp was executed. The location of the sensors on the objects was chosen by checking the most frequent contact points between the hand and the object through the ContactDB database [28] and by direct observation.

The force was acquired by means of a Printed Circuit Board (PCB) connected to a National Instruments DAQ USB-6210. The DAQ sent voltage data from the sensors to a computer in which a custom-made Graphical User Interface developed in C# received and saved voltage data at 100 Hz. The GUI allowed starting and stopping the acquisition and sending an analogical trigger to the other acquisition systems in the experimental setup. Finally, from the GUI it was also possible to send sub-triggers that allowed segmenting data into the reaching and static phase of the reach-to-grasp tasks.

2.1.4. Muscle analysis

Muscular activity was recorded at 1000 Hz using surface electromyography (sEMG). 7 wireless Delsys Trigno sensors were positioned on 7 selected muscles following the SENIAM guidelines [29] (see Fig. 1A - Muscular Activity). The monitored extrinsic muscles were: *Flexor Carpi Radialis* (FCR), *Flexor Carpi Ulnaris* (FCU), *Extensor Carpi Radialis* (ECR), *Extensor Carpi Ulnaris* (ECU), *Flexor Digitorum Communis* (FDC), *Extensor Digitorum Communis* (EDC), *Abductor Pollicis Lungus* (APL). The choice of the muscles was made by selecting the ones monitored in most works about hand muscular synergies. In addition, we chose to monitor only extrinsic muscles as the hand dorsum was already covered with markers and it was not possible to position also sEMG electrodes to monitor intrinsic muscles. Moreover, it was not possible to position electrodes on the palm of the hand as the task included the interaction with objects which could have introduced artifacts in the signal.

2.2. Experimental protocol

Eight healthy volunteers participated to the experimental acquisitions. The characteristics of the enrolled volunteers are resumed in Table 2. All subjects were right-dominant. Attention was paid in recruiting participants with different hand lengths, measured from the distal wrist crease to the tip of the middle finger, to guarantee the heterogeneity of the population.

Table 2
Characteristics of the participants.

Volunteer	Age	Hand length [mm]	Sex
1	28	180	F
2	25	170	F
3	28	210	M
4	27	182	F
5	26	185	M
6	27	180	F
7	25	180	F
8	26	200	M

Before performing the experimental trials, the procedure of the acquisitions was clearly explained to the participants. All of them provided written informed consent. The experimental protocol was approved by the local Ethical Committee (Comitato Etico Università Campus Bio-Medico di Roma, reference number: N. 41/17 OSS ComEt CBM) and complied with the Declaration of Helsinki.

The volunteers were required to perform reach-to-grasp tasks of the selected objects, for a total amount of 21 grasps. The order of the grasps to be executed was randomized for each participant.

The experimental acquisition was divided in four phases:

1. Positioning of the markers and sEMG sensors on the volunteer (~10–15 min).
2. Recording of calibration movements with the BTS to compute the hand joints Ranges of Motion (RoMs) to normalize kinematic data in post-processing. The participants sat comfortably on a chair, with the elbow positioned on the table at 90° and were instructed to execute FE and AA movements of all finger joints and FE and RUD movements of the wrist to their full RoM (~ 10 min).
3. Reach-to-grasp tasks of the provided objects. The volunteers performed one repetition of each grasp. The trial started with the participant comfortably sitting on the chair in front of the table, as shown in Fig. 1A, and the markerised hand positioned in a open-hand posture on the table. At the start provided by the experimenter, the volunteer performed the reaching task to grasp as naturally as possible, with the markerised hand, the object positioned on the table. He/she was asked to pay attention in making contact with all the sensors positioned on the object. When a stable grasp was achieved by the participant, a sub-trigger was sent from the force GUI to segment the reaching from the static grasp phase. Then, the subjects kept the static grasp posture for 5 s. The acquisition stopped at the end of the static phase. The reach-to-grasp task lasted about 90 min. The volunteers could rest according to their needs between the tasks to avoid fatigue. The entire experimental acquisition lasted about 2 h per subject.

During the static phase of the grasping tasks, subjects were not asked to modulate the force, but to exert a stable grasping force associated to a stable muscular activity. Therefore, the analysis conducted in this work is focused on the static grasping configuration of the hand in terms of kinematics, exerted forces and muscular activity. In particular, muscular activity is here studied in relation to the contraction state of the muscles when holding the objects in a stable manner.

2.3. Data analysis

2.3.1. Pre-processing

Joint angles were computed from marker coordinates by implementing in the BTS software the kinematic protocol proposed by Cordella et al. in [26], with the integration of the estimation of the wrist angles as described in [27]. Then, joint angle data were exported to text

files to be processed in MATLAB R2021b (The MathWorks, Inc.). They were filtered with a 4th order low pass Butterworth filter with cutoff frequency 1.5 Hz, chosen by applying the residuals method [30], and visually inspected trial by trial. Joint angles were normalized with respect to the subject-specific RoM of each joint [8].

Similarly, voltage data recorded from the force sensors were imported in MATLAB and converted in force values. Contact forces exerted by each finger and palm areas were normalized, for each subject, by the maximum force exerted by each district among all the trials.

Finally, sEMG data were imported in MATLAB and filtered with a 4th order bandpass Butterworth filter with cutoff frequencies 20–450 Hz and a 4th order stopband Butterworth filter to remove the 50 Hz frequency. Then, data were rectified and normalized with respect to the maximum contraction of each muscle for each subject. For the purpose of this work, only the static grasp phase of the trial was considered for extracting postural, force and muscle synergies.

2.3.2. Dimensionality reduction

In this work, the expectation maximization (EM)-PCA proposed in [24] was exploited as a dimensionality reduction technique to extract postural, force and muscle synergies. EM-PCA allows performing PCA on datasets with missing data or data affected by measurement noise, and retrieving a set of orthogonal eigenvectors that are optimized to describe most of the data variance with the lowest number of vectors. The eigenvectors represent the directions of maximum data variance, and they are ordered in descending percentage of explained variance. These eigenvectors will be hereafter also called *loading vectors*. The measurement error is estimated and used to weight the input data, so that the resulting eigenvectors are able to describe the true underlying signal variation.

Given a matrix $\mathbf{X}$ of dimension $m \times n$, where m are the data samples and n the variables (e.g. the angles, contact points and muscles), the algorithm implemented in MATLAB R2021b and based on Singular Value Decomposition, solved the following minimization problem which allowed calculating the optimal eigenvectors by applying the EM iterative method [24]

$$\chi^2 = \sum_{i,j} \mathbf{W}_{ij} \left[\mathbf{X} - \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T\right]_{ij} \quad (1)$$

where i and j represented the rows and columns respectively, $\mathbf{W}$ $(m \times n)$ was the weights matrix, $\mathbf{U}$ $(m \times n)$ was the matrix of the left singular vectors, $\boldsymbol{\Sigma}$ $(n \times n)$ was the diagonal matrix of singular values, $\mathbf{U} * \boldsymbol{\Sigma}$ represented the data in the new synergy space, and $\mathbf{V}$ $(n \times n)$ was the matrix of the right singular vectors (or *loadings* matrix), i.e. the optimal eigenvectors. The variance explained by the ith eigenvector could be computed from the singular values as $variance(i) = \sigma_i / \sum_i \sigma_i$, where σ_i is the ith singular value.

To ensure that all the trials had the same importance in the analysis, the static phase of the grasp for each trial and each subject was resampled to 300 frames for kinematic and force data, and to 5000 frames for EMG data. Then, the data matrix on which to apply EM-PCA was constructed by appending all the static phase trials.

2.3.3. Grasp classes selection

The first type of analysis conducted in this work required the identification of the classes on which to further extract subject-specific and task-specific synergies. In fact, we hypothesize that synergies extracted on grasp classes that have different features could be more representative of the movement associated to the different grasping strategies, e.g., shaping the hand to perform a power or a precision grasp by involving or not the palm in the grasp. Therefore, we investigated whether different grasp classes could be distinguished in the synergy space, after extracting synergies on data from all subjects and all trials together, similarly to the analysis conducted in [18]. On the basis of the grasp taxonomy provided by Cutkosky [23], grasps were labeled as power and precision grasps (2 classes), i.e. the two major grasp classes that include grasps that involve or not the palm in grasping task, respectively, and as power prismatic, power circular, precision prismatic and precision circular grasps (4 classes), i.e. the four "sub-classes" identified by Cutkosky depending on the symmetry of the fingers around the object. It is important to underline that power and precision grasp classes include in turn different types of grasps that can be performed with different objects. Similarly, the "sub-classes" circular and prismatic include different types of grasps, such as pinch, lateral, tridigital and pentadigital, therefore the grasps included into the same class or sub-class can be different among each other in terms of hand posture (i.e., angular configuration) but are classified into the same branch because they share some features, e.g. the involvement of the palm in the grasp, the symmetry of the finger configuration or the position of the thumb. EM-PCA was applied to normalized kinematic (i.e., joint angles), contact force and EMG data separately by appending in the $\mathbf{X}$ data matrix all trials from all the subjects, with the aim of representing data in the new synergy space of uncorrelated variables. To check to what extent grasp classes could be distinguished in the synergy space, a linear classifier was applied due to its speed and simplicity. A previous work [18] used a linear classifier to investigate whether grasps could be separated by a set of 3D planes in the synergy space, and highlighted that grasps could be distinguished both in the kinematic and muscle domain. In this work, Linear Discriminant Analysis (LDA) was applied to classify data in the synergy space into the selected grasp classes by using the Leave One Subject Out (LOSO) technique: the LDA classifier was trained on data from 7 participants and tested on data from one subject, by varying each time the subject to be left out of the training set for each iteration of the classification. The average accuracy of the LDA classifier (i.e. number of grasps correctly identified with respect to the total of grasps) was then computed on the 8 repetitions as an index of the capability to discriminate grasp types in the PC space starting from kinematic, force, and muscle data, for each grasp classification (2 classes or 4 classes). The same type of analysis was also conducted by considering combined (kinematic and force, kinematic and EMG, force and EMG and kinematic, force and EMG) data organized in the matrix on which to compute PCA. This allowed understanding whether the inclusion of additional information such as grasp force and/or muscle activation allowed a better classification of data in the PC space.

2.3.4. Subject-specific and task-specific synergies

Once the grasp classes on which to extract subject-specific synergies were identified, EM-PCA was applied, for each grasp class and subject per subject, on normalized kinematic (i.e., joint angles), contact force and EMG data and on combinations of such data. As previously described, both "pure" kinematic, force and muscle datasets as well as "combined" datasets were analyzed. In particular, combined datasets were generated by including in the $\mathbf{X}$ matrix different types of data, e.g. joint angle data for the first 23 columns and contact force data for the last 7 columns for combined kinematic-force synergies extraction. The application of EM-PCA on the data matrix with combined data was performed to reveal possible relations between different variables involved in the grasping action. In this work, synergy extraction was performed separately for each subject to avoid the risk of extracting "unreal" synergies, as different individual motion strategies could generate unreal global synergies [8].

The similarity between the computed subject-specific synergies was evaluated by means of the Cosine Similarity (CS) index. This index was previously used in literature to quantify similarity between the kinematic synergies of different locomotion tasks [31], hand kinematic synergies during environmental constraint exploitation [7] and between muscle synergies among subjects and sessions [32–36]. The CS index computes the similarity between two loading vectors (or synergies) in absolute value, and it ranges from 0 to 1, which corresponds to no similarity and high similarity, respectively. Based on

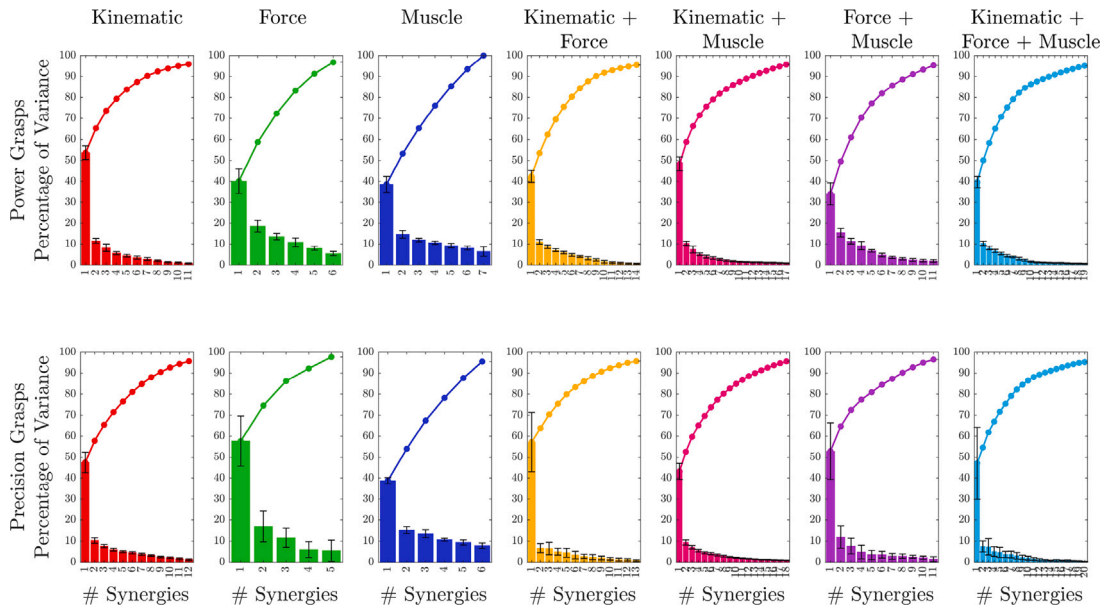


Fig. 2. Average percentage of variance and standard deviation expressed by each PC for subject-specific synergies. For each combination of synergies, the percentage of variance expressed by each PC and the cumulative sum of the variance up to the number of components that express 95% of variance are shown, for power and precision grasps.

previous works, a threshold of 0.8 was chosen to indicate highly similar synergies [31,32].

Differently from the literature works that extracted average generalized synergies after applying a hierarchical cluster analysis [8,10,11], the use of the CS index to compare synergies allows taking into account the individual motion strategy of each participant also in terms of variance. In fact, the hierarchical clustering merges synergies not considering their hierarchy in terms of variance, i.e. it could merge the first, dominant synergy of one subject with the third high-order synergy of another, even though the importance of those synergies is different in terms of expressed variance. Instead, the use of the CS index allows comparing synergies of the same order, and this allows quantifying how much the principal grasp strategies are similar between subjects.

Before proceeding with the calculation of the CS index, a statistical analysis was conducted to verify the existence of a statistical difference between subject-specific synergies of the same order [36]. Average CS indices for synergy 1 to n, where n is the number of synergies extracted in each domain, were then extracted by first calculating the similarity between same-order synergies of pairs of subjects, and then averaging the indices computed for all subject pairs. In addition, to verify whether the similarity of subject-specific synergies was significantly higher than the random matching, the average similarity of any two paired random vectors of the matrix containing the synergies extracted for all subjects as columns, for each domain, was computed, similar to the analysis conducted in [36]. After computing the average CS index on the 8 volunteers for each same-order synergy, the average synergies were extracted as the mean of the absolute values of each loading vector. The absolute values of the loadings was used for computing the CS indices as well as for extracting the average synergies, as the signs of all loadings are arbitrary and only the analysis of their relative magnitude is meaningful for understanding the most important variables for each synergy [37,38]. This allowed analyzing the generalized grasp strategies in terms of kinematics, force and muscular activity, and investigating possible relations between kinematic, force and muscle domains.

The posture of the hand, obtained by activating each kinematic synergy, was visualized by means of a virtual hand developed taking advantage of the SynGrasp [39] MATLAB toolbox. In particular, a 21 DoFs virtual hand was implemented in this toolbox [27]. It took as input the marker positions of one representative volunteer to define the hand joint reference frames. The hand posture was generated by imposing a synergy activation coefficient of +1 and −1 to visualize the corresponding closed (flexed) and open (extended) configuration of the hand generated by activating a synergy, according to the equation $q = \mathbf{S} \cdot z$, where $q \in R^{21}$ is the angular configuration of the hand joints, $\mathbf{S} \in R^{21\times21}$ is the synergy matrix (i.e. the matrix of the *loadings*) and $z \in R^{21}$ is the synergy activation vector. The wrist DoFs were not represented as the virtual hand only allowed defining the fingers and palm.

Finally, simulations were conducted in Simulink and Simscape Multibody [40] to evaluate the stability of grasps generated by activating the previously computed synergies. The 21 DoFs hand model was used to simulate power and pentadigital precision grasps of a sphere with a diameter of $4 \cdot 10^{-2}$ m [41,42]. The object was positioned with its center of mass aligned with the middle finger MCP joint [26], and fixed to the world space through a weld joint. To evaluate grasp stability with different initial object positions, the object orthogonal distance from the hand palm was varied from $48 \cdot 10^{-2}$ m to $56 \cdot 10^{-2}$ m for power grasps and from $68 \cdot 10^{-2}$ m to $76 \cdot 10^{-2}$ m for precision grasps, with $2 \cdot 10^{-2}$ m steps. The hand joints were moved of fixed angular steps chosen by computing the angular displacement consequent to the activation of the first two (power or precision) synergies (i.e., the synergies with highest similarity among subjects) with $\Delta z = 0.1$, according to $\Delta q = \mathbf{S} \cdot \Delta z$. In particular, for generating the movement with combined kinematic and force synergies, only the *loadings* corresponding to the first 21 elements of each synergy, i.e., the kinematic part of each synergy, were considered. The initial hand posture corresponded to a relaxed hand posture. The index chosen to evaluate grasp stability is the *Posture of Hand Finger Joints* [27,43]. It measures how far each joint is from its maximum and minimum RoM during the static grasp phase, and its value goes from 0 (unstable grasp) to 1 (stable grasp).

3. Results

Table 3 shows the mean and standard deviation of the accuracy of the LDA classifier computed on 8 repetitions of the classification by exploiting the LOSO technique. As it will be further explained in Section 4, based on these results and on the search for a trade-off between classifier accuracy and the number of grasps belonging to each class, the extraction of synergies was conducted on power and precision grasps separately.

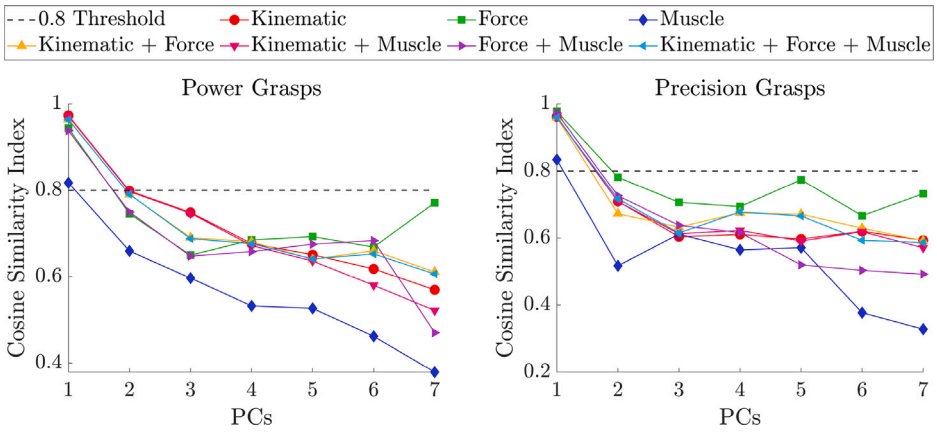


Fig. 3. Average Cosine Similarity Index computed between each same-order synergy on the 8 volunteers.

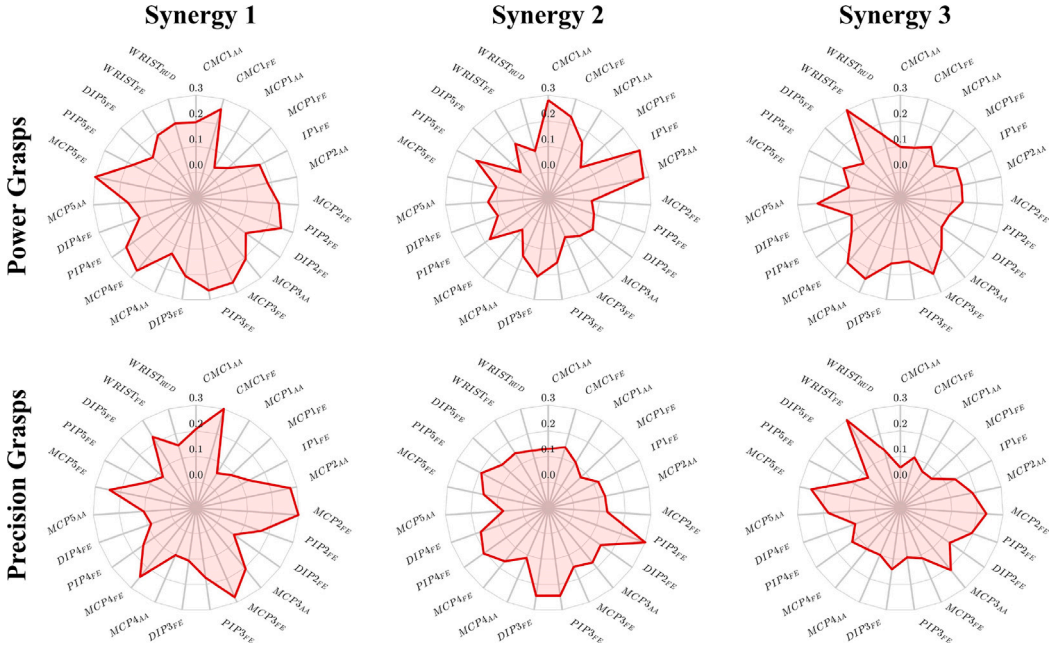


Fig. 4. Average kinematic synergies for power and precision grasps: the contribution of each variable, i.e., joint angle excursion, to the synergy is shown in a polar plot.

Table 3
Results about the accuracy of LDA.

	Power/Precision (Mean ± SD)	Power Prismatic/ Power Circular/ Precision Prismatic/ Precision Circular (Mean ± SD)
Kinematic	0.89 ± 0.08	0.74 ± 0.08
Force	0.77 ± 0.08	0.49 ± 0.08
Muscle	0.66 ± 0.08	0.45 ± 0.12
Kinematic + Force	0.89 ± 0.08	0.73 ± 0.11
Muscle + Force	0.89 ± 0.08	0.75 ± 0.06
Muscle + Kinematic	0.89 ± 0.08	0.75 ± 0.06
Kinematic + Force + Muscle	0.89 ± 0.08	0.75 ± 0.06

The variance expressed by the computed synergies is reported in Fig. 2: the percentage of variance expressed by each PC and the cumulative sum of the variance up to the number of components that express 95% of variance are shown for each combination of synergies. It is evident that the first synergy allows explaining a high percentage of the data variance in all domains.

The non-parametric Kruskal–Wallis test was applied due to data distribution, assessed through the Shapiro–Wilk test, to verify the existence of a statistical difference between subject-specific synergies of the same-order. Results showed that synergies were not different among each other in a statistical significant manner (p-value > 0.05) in all domains, thus paving the way for a CS analysis. Fig. 3 shows the results of the CS analysis performed between subject-specific synergies. In particular, synergies extracted on "pure" and "combined" datasets are shown and results of the average CS indices calculated on the 8 volunteers for the computed synergies up to 7 PCs are reported for brevity. Random similarity values were computed as in [36]. Median random similarity values were $[0.61, 0.73, 0.57, 0.57, 0.49, 0.37, 0.49]$ for power grasps and $[0.61, 0.73, 0.53, 0.62, 0.49, 0.44, 0.50]$ for precision grasps, where values in the arrays refer to kinematic, force, muscle and combined kinematic-force, kinematic-muscle, force-muscle and kinematic-force-muscle synergies, respectively. CS indices were compared to the random similarity values by applying the Kruskal–Wallis test. The first synergies had CS indices statistically higher than the random similarity in all domains ($p - value < 0.05$). In particular, in the kinematic domain, synergies 1–4 and 1–2 for power and precision grasps, respectively, had CS indices statistically higher than random similarity. Combined force and kinematic synergies extracted on power

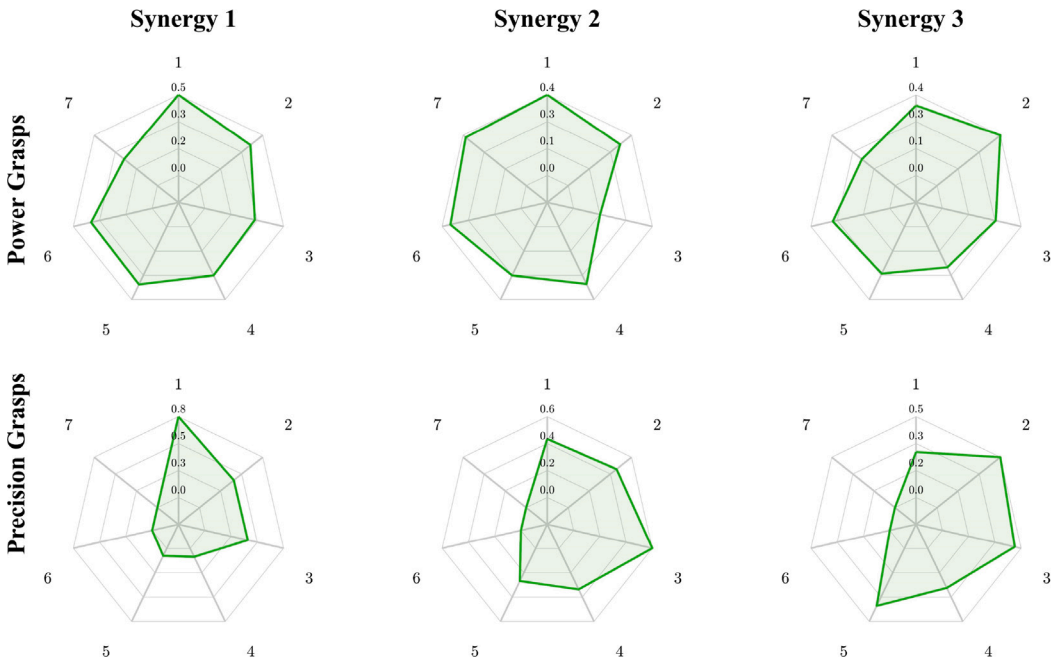


Fig. 5. Average force synergies for power and precision grasps: the contribution of each variable, i.e., contact force exerted on the contact point identified with number 1 to 7, to the synergy is shown in a polar plot.

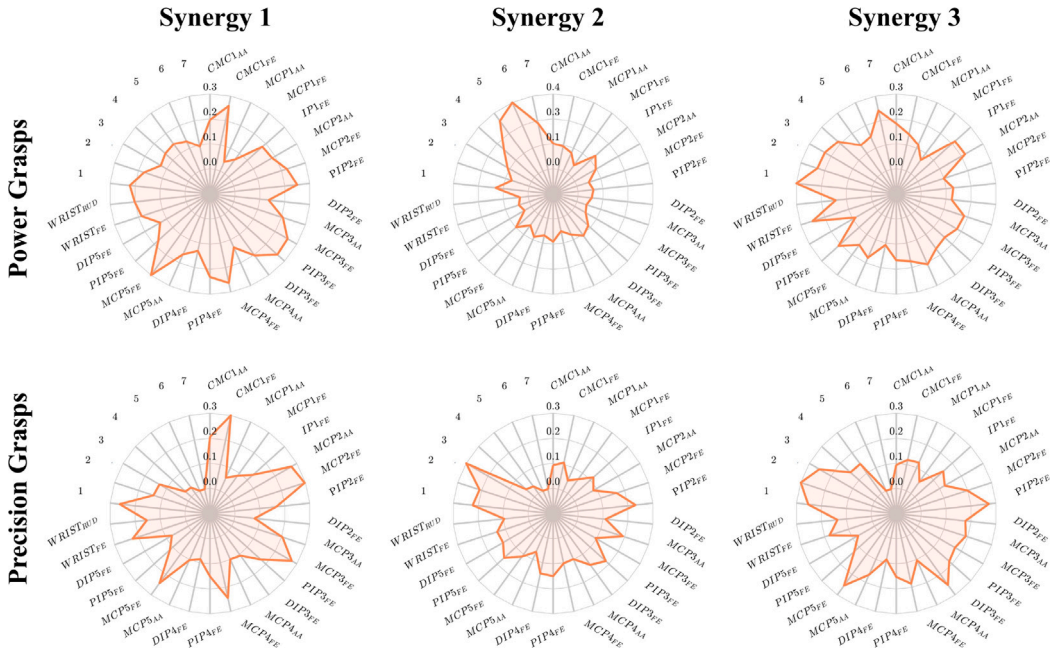


Fig. 6. Average kinematic and force synergies for power and precision grasps: the contribution of each variable, i.e., joint angle excursion and contact force, to the synergy is shown in a polar plot.

grasps had CS indices statistically higher than random similarity for synergies 1–6. Finally, synergies extracted on combined kinematic, force and muscle domains had CS indices statistically higher than random similarity for synergies 1–4 to 1–7.

Based on the results of the CS analysis, average synergies were computed to represent the general kinematic, force and muscular patterns, as well as combined patterns. Figs. 4–6 show the contribution of each variable to the correspondent synergy. For the sake of brevity, only the most significant results are graphically reported, i.e. the average kinematic, force and combined kinematic and force synergies.

The analysis of muscle synergies, in fact, revealed that all muscles contributed with approximately the same weight to the synergy, consistently with the results obtained in [18], and the first, second and third synergies did not differ particularly among each other. The most evident difference in muscle synergies between the two types of grasps consisted in the contribution of the FCR muscle, which was more marked in power with respect to precision grasps for all the first three synergies. In addition, the contribution of the APL became progressively less important in higher order synergies for precision grasps. Given the non distinctive pattern of muscle synergies, results about combined datasets with EMG data were not explicitly reported in this paper because it was not possible to identify a characteristic contribution of muscle data to the synergies. What emerged, instead, was that in

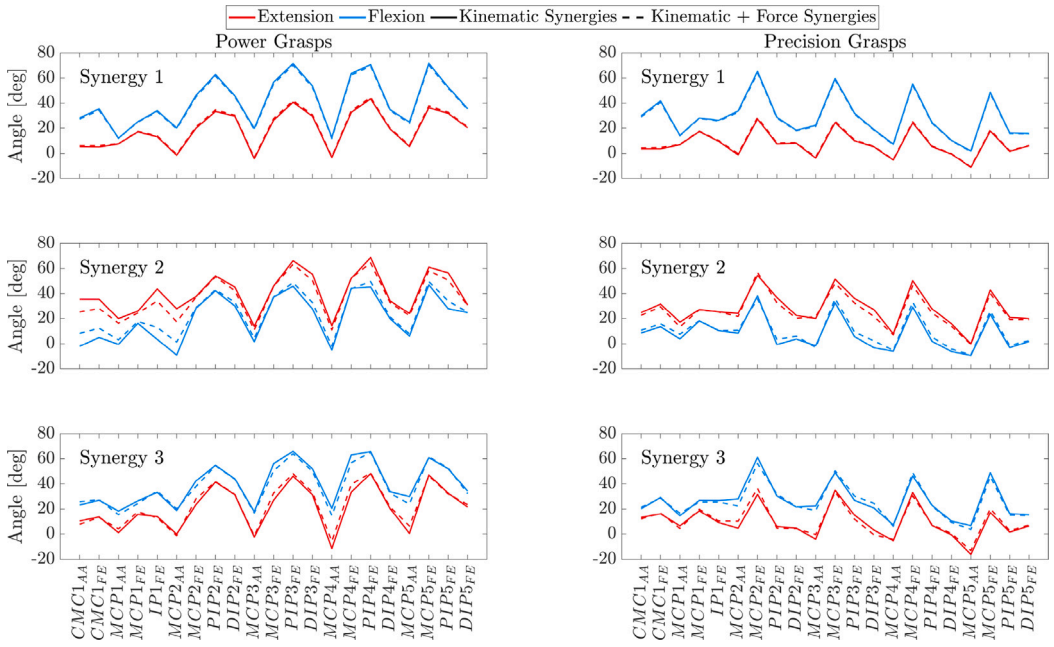


Fig. 7. Difference of angular configuration adopted by the virtual hands by activating the first three synergies extracted on the kinematic dataset (continuous line) and on the kinematic + force dataset (dashed line). The extended and flexed hand configuration are shown in red and blue respectively.

synergies extracted from the combined dataset, forces always played a major role in each synergy with respect to EMG data.

Fig. 7 reports the angular configuration adopted by the virtual hands by activating kinematic synergies or combined kinematic-force synergies. In particular, for the kinematic-force synergies, only the loadings of the kinematic domain, i.e. the first 21 items of each synergy re-tuned by the contribution of contact forces, were considered to generate the correspondent hand posture. Only the posture associated to the activation of the first three synergies is reported for brevity. The configuration of the hand was obtained by imposing values of the synergy activation vector of −1 and +1 for the extended and flexed configuration, respectively.

Results about the simulations conducted in Simulink and Simscape Multibody showed that, for both power and precision grasps, stability increased when hands were activated with combined kinematic and force synergies rather than with kinematic synergies only. In particular, the Kruskal–Wallis test was conducted to evaluate whether the *Posture of Hand Finger Joints* indicators were statistically different between the two movement strategies (i.e., activation of pure kinematic or combined kinematic and force synergies). Median stability values for power grasps were 0.64 and 0.89 (p-value = 0.009) for kinematic and combined kinematic and force synergies, respectively. For precision grasps, median stability values were 0.86 and 0.90 (p-value = 0.009) for kinematic and combined kinematic and force synergies, respectively. Therefore, stability significantly increased when generating grasps through the activation of combined kinematic and force synergies.

Finally, Fig. 8 reports the posture of the virtual hands activated with the first two generalized kinematic synergies, shown in the extended (min) and flexed (max) configuration. Hand postures were generated by imposing a synergy activation vector value of −1 and +1, respectively. The average posture of the hand is also reported in the middle of each figure.

4. Discussion

A preliminary analysis was performed to identify the most suitable dimensionality reduction technique to be applied to extract synergies in this work. Both linear and non-linear methods were proposed in the literature to extract synergies in different domains. In particular, linear methods such as PCA and NNMF were mostly used to extract motor synergies [2,3] due to their ease of interpretation and limited computational burden [38]. Non-linear methods were also explored to characterize hand kinematics for prosthetic hands control applications [4], obtaining promising results compared to linear methods. A previous work by Lambert-Shirzad and Van der Loos [2] compared different linear matrix factorization methods, including PCA and NNMF, on real experimental data, and concluded that PCA and NNMF had comparable performance on both kinematic and EMG data. Among linear methods, NNMF is a widely exploited technique to extract and analyze muscle synergies. It allows extracting activation patterns of the muscles in the time domain and movement primitives that can simplify the description of the whole movement generation in the muscle domain [33]. Since this work was interested in extracting synergies only in the static grasp phase, we decided to neglect the analysis in the time domain and to reduce the problem complexity. Therefore, PCA was selected as the dimensionality reduction technique to extract synergies in this work, also due to its extremely widespread use in this field. Lastly, since the objective of this work was to extract synergies on combined datasets (i.e., kinematic, force and muscle data), there was the need to use the same dimensionality reduction technique for all domains, to compare the results. The choice of using PCA to extract synergies in all domains, with respect to non-linear methods, also allowed comparing the obtained results with previous similar works that used PCA to extract kinematic, muscle and force synergies. Following these reasons, the PCA and, in particular, EM-PCA was selected as the dimensionality reduction technique to extract synergies.

As evident from Table 3, the subdivision into power and precision grasps allowed reaching 89% accuracy of a LDA classifier for kinematic data. This means that grasps that are classified in the Cutkosky grasp taxonomy as power and precision grasps can be clearly distinguished in the synergy space because they exhibit distinctive features, even though grasps belonging to the same class can be different in terms of hand posture. The accuracy in discriminating power and precision grasps computed on force data was only 77%, and decreased to 66% for EMG data. This was consistent with previous literature findings [18], in which the authors showed that the accuracy of a linear classifier in separating grasp types in the PC space, evaluated by means of the

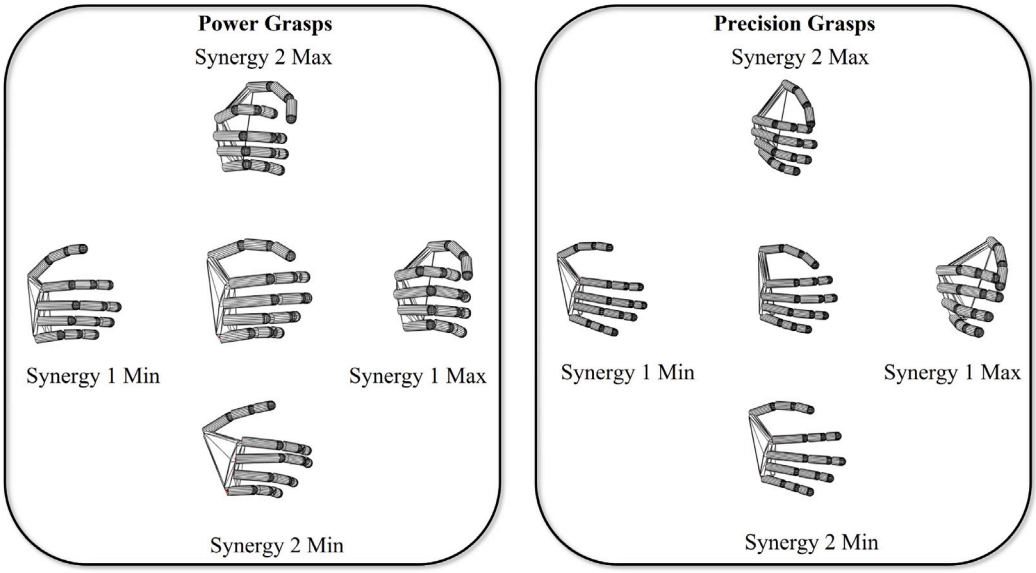


Fig. 8. Kinematic synergies obtained for the power and precision grasps. The virtual hands activated with the first two synergies are shown in the extended (min) and flexed (max) configuration.

Balanced Error Rate, was higher for kinematic data than for EMG data. Moreover, the combined datasets produced the same accuracy (89%) in classifying data between power and precision grasps. Accuracy of the LDA classifier was also high for the four classes classification (i.e., about 75% for kinematic and combined datasets and about 45%–49% for muscle and force data, respectively), however, if four classes were selected to extract synergies, there would have been only three similar grasps belonging to the power circular class, thus not guaranteeing enough variety of grasps on which to extract synergies. Hence, a compromise was reached between achieving high accuracy of the LDA classifier and allow extracting synergies from a grasp class that encompassed sufficiently large and heterogeneous number of grasps. Task-specific synergies were therefore extracted on power and precision grasp classes.

The obtained results seem to be opposed to the results presented in [6], where the authors pointed out that the hand postures representing power and precision grasps did not tend to cluster into discrete groups. However, this apparent difference in the results could be probably due to the fact that the hand postures considered in [6] were imaginary postures, so there was a lack of mechanical contact and somatosensory feedback which can influence the hand posture itself [12,44].

The analysis of the variance explained by each eigenvector (Fig. 2) highlighted that the EM-PCA algorithm allowed determining an optimized predominant direction in the synergy space which expressed most of the data variance, and this was true for all the computed synergies. In fact, the first eigenvector alone explained, on average, around 50% of variance for kinematic synergies, 40% and 58% of the variance for force synergies for power and precision grasps, respectively, and around 40% of the variance for muscle synergies. This means that the majority of the information content is represented by the first component. The lower variance expressed by the extracted kinematic synergies compared to the work by Santello et al. [6] in which the first two PCs accounted for 80% of variance, could be explained by the fact that in [6] static postures were recorded while grasping imaginary objects. As also hypothesized in [12,44], the contact or somatosensory feedback could alter the computed synergies, and a higher number of principal components could be needed to reconstruct the hand posture when interacting with real objects. Therefore, the lower variance explained by the first principal components in our work could be related to the contact with the objects. Nevertheless, the percentage of variance explained by the first three synergies in our work is similar to the results obtained by Liu et al. [45], Patel et al. [19] and Thakur et al. [46] for kinematic synergies, who obtained 65%, 75% and 80% of variance explained by the first three PCs, respectively [10]. All these works used real objects to conduct the experiments. Jarrassè et al. [12] and Mason et al. [47] also used real objects and obtained 93% and 99% of variance explained by the first three PCs. However, they analyzed a reduced set of grasps (i.e., 9 and 5, respectively). The reduced variety of the considered tasks could have influenced the results on the variance, reducing the number of PCs needed to reconstruct original data. For force synergies, the percentage of variance explained the first three PCs in this work (i.e., 72% and 86% for power and precision grasps, respectively) is higher compared to the results obtained by Starke et al. [15], who obtained less than 70% of variance explained by the first three components, thus suggesting that the application of EM-PCA allowed extracting optimized eigenvectors able to describe most of the data variance with a lower number of vectors. Finally, results on muscle synergies obtained in our work were similar to the results obtained by Weiss and Flanders [16], where the first PC explained about 40% of variance and 3–4 PCs were needed to explain more than 80% of the variance. Combined datasets showed the same behavior of pure datasets, even though the number of components to reach 95% of the variance highly increased with respect to "pure" kinematic, force and muscle synergies. This could be related to the larger amount of variables considered in the analysis to describe the motor task. For all the analyzed synergies, with 2–5 PCs it was possible to represent ~ 70% of the variance.

Results about the CS analysis (see Fig. 3) showed that the first synergies extracted for all the possible data combinations were very similar between subjects ($CS > 0.8$), for both power and precision grasps. This is a very important finding, as it suggest that the principal kinematic, force and muscle patterns adopted to perform specific grasping tasks are widely shared among healthy volunteers, and this allows defining generalized grasping strategies in the three domains. The first synergies, that explained most of the variance of the original data, could be therefore interpreted as a general pattern shared among different subjects. This is true for synergies extracted in all the three analyzed domains. Moreover, these patterns were distinctive for power and precision grasps, and this was more evident for kinematic and force synergies. Higher order synergies, instead, seemed to codify for more subject-specific grasp strategies that were not widely shared among

subjects, as supported by the results of the CS. In fact, similarity tended to decrease for higher order synergies. For force synergies, instead, an increase of similarity was appreciable for higher order PCs. This suggested that, in addition to a general force pattern represented by the first force synergy shared among subjects, the grasp action could be regulated by force adjustments represented by higher order synergies that were also shared among volunteers.

To verify this aspect, the forces exerted by each subject were recalculated by activating each force synergy one at time. For power grasps and for all the volunteers, average normalized forces were around 0.5 for each contact point for the first synergy and below 0.2 for higher order synergies. This suggests that higher order synergies may codify for very small force adjustments adopted to obtain a stable grasp. For precision grasps, instead, force variations up to 0.5 could be observed on the five fingers for the first and the fifth, sixth and seventh synergy. For the other synergies, instead, exerted forces were around zero. This suggests that the force adjustments performed in precision grasps and detectable in higher order synergies are more important than the force adjustments performed in power grasps. Precision grasps, in fact, require an optimal force distribution among fingers to hold the object firmly, given the lower number of contact points on fingers. The forces exerted in both power and precision grasps recalculated by activating each synergy were compared among subjects by applying the Kruskal–Wallis test. Results revealed that force magnitudes did not differ among subjects in a statistically significant manner, therefore the magnitude of the force variations to adjust grasps was shared among subjects, as confirmed by the CS indexes that were around 0.7 for synergies 1–7 for both power and precision grasps.

To verify whether postural synergies extracted from all subjects together were actually different with respect to average synergies computed from subject-specific ones, as stated in [8], task-specific kinematic synergies were also extracted from a matrix containing data from all subjects together. What emerged was that the higher was the CS index computed on subject-specific synergies, the more the average synergy and the synergy extracted considering all subjects together were similar. This means that the risk of retrieving “unreal” synergies, which exists when extracting synergies on data from all subjects together, is confined to higher order synergies that represent more subject-specific characteristic motion patterns. In fact, the computation of the CS index between each same-order synergy calculated in the two cases (i.e., average synergies and synergies extracted from the data matrix with all subjects) revealed that first order synergies were identical for both power and precision grasps ($CS = 1$), and similarity decreased for higher order synergies (i.e., 0.97, 0.93 and 0.71 for the second, third and fourth synergy of power grasps and 0.93, 0.83 and 0.83 for the second, third and fourth synergy of precision grasps).

Fig. 4 shows the average kinematic synergies computed for power and precision grasps: the contribution of each variable to the synergy is appreciable in the polar plot. The first three synergies were only reported for brevity, as they are able to describe most of the variance of the dataset ($\sim 70\%$). As evident, synergies were distinctive for the two types of grasps. For kinematic synergies extracted on power grasps, a major contribution to the first synergy was given by the FE and AA of the long fingers and CMC1 joints. This activation pattern corresponded to a general open-closing pattern of the hand, which can be visualized also in Fig. 8 (Power Grasps). The wrist also contributed to the synergy with both DoFs. The second synergy was characterized by the main contribution of the CMC1 AA and MCP2 AA, which corresponded to the adjustment of the grasp aperture, i.e. the aperture angle between the thumb and the index finger, as well as by the FE of the IP1 joint. Finally, the third synergy showed a major contribution of the middle, ring, little finger and wrist FE. The first kinematic synergy extracted on precision grasps showed a minor contribution of PIP and DIP joints with respect to power grasps. In fact, as also shown in Fig. 8 (Precision Grasps), the first synergy corresponded to a general open-closing pattern of the hand which, differently from power grasps, was performed with the PIP and DIP joints more extended to perform precision grasps. The AA was pronounced mainly at the thumb, index and middle finger. The second synergy showed a major importance of the FE of the PIP and DIP joints, which contribute to increase the opening–closing of the hand, whereas in the third synergy the index and little finger and wrist FE were activated.

Average force synergies, reported in Fig. 5, were also distinctive for power and precision grasps. In fact, for power grasps there was quite uniform contribution of the force exerted at the fingertips contact points (1–5), followed by the contribution of the palm (6–7). The second and third synergies were quite similar among each other. For precision grasps, instead, synergies highly differed as the order increased: the first synergy was characterized by the activation of the contact points at the thumb, index and middle fingertips. This means that the first three fingers are the most used during precision grasp tasks. Moreover, it is important to highlight how the loading assigned to the thumb was the highest and it was approximately twice the loading of the index and middle finger. This is consistent to the fact that the normal force exerted by the thumb is equal and opposite to the sum of the forces exerted by the long fingers involved in the grasp to satisfy the equilibrium, even though the distribution of the force among the long fingers is indeterminate [22]. The remaining fingers and, in particular, the palm, did not contribute to the synergy as they were less or at all activated. In the second and third synergy, the contribute of the other long fingers increased while the contribution of the thumb decreased. Therefore, higher order synergies seem to account for a subject-specific fine distribution of the force among the long fingers to generate a stable grasp.

The analysis of the combined kinematic and force data (Fig. 6) revealed how forces tended to progressively acquire higher importance from the first to higher order synergies. In fact, in the first synergy of both power and precision grasps, the contribution of force data approximately equaled the one of kinematic data, with the same pattern showed for “pure” synergies. For higher order synergies, instead, force contribution became more important. This suggests that the regulation of hand posture may take into account an additional contribute given by the force distribution, that actually modifies the hand posture itself. This is a very important finding as it appears to be opposite to the hypothesis proposed by Santello et al. in [6] that the regulation of hand posture was separated from the regulation of the contact forces. Our results, instead, show that as the synergy order increases, forces start playing an important role in generating the optimal hand posture for performing a particular grasp type by slightly changing the angular configuration of the joints. This finding was also confirmed by the results shown in Fig. 7: the angular configuration adopted by the virtual hands by activating kinematic synergies only (continuous line) or combined kinematic-force synergies changes as the synergy order increases. In other words, it seems that forces contribute to the generation of the hand posture by re-tuning the angular configuration according to the optimal force pattern to be applied to generate a stable grasp. To verify whether stability increased by activating combined kinematic and force synergies rather than pure kinematic synergies only, simulations were conducted in Simulink and Simscape Multibody. Results showed that for both power and precision grasps, grasp stability, calculated through the *Posture of Hand Finger Joints* indicator [27], significantly increased when grasps were generated though the activation of the first two combined kinematic and force synergies, compared to kinematic synergies only. This result supports the hypothesis that the regulation of hand posture is not independent from the regulation of contact force, and force synergies may re-tune the angular configuration to improve grasp stability.

The extracted muscle synergies were not distinctive for the specific grasp subdivision considered in this work. In general, all muscles contributed almost equally to each synergy. A similar result was obtained in [18], where a co-activation of all muscles characterized the first muscle synergy extracted on muscle data regarding grasping tasks of

five different objects. In addition, both Weiss and Flanders [16] and Castellini et al. [18] did not found a characteristic correspondence between muscle and postural synergies, as different muscle groups could be activated to generate similar hand postures. Therefore, it is not possible to associate the activation of a kinematic or combined kinematic-force synergy based on the activation of a particular muscle synergy.

4.1. Limitations of the study

The analysis performed in this work is subject to possible limitations related to the reliability of sEMG for measuring muscle signals and of piezoresistive force sensors for recording grasping forces. In fact, the sEMG technique is associated to several problems such as cross-talk, sweating, possible inaccurate electrode positioning, etc. [18]. Despite its well-known limitations in monitoring muscular activity, sEMG is also cheap, non invasive and easy to use and was exploited in several works to extract muscle synergies [16–18,48]. In this work, we positioned sEMG electrodes carefully preparing the skin and following the SENIAM guidelines. In addition, the monitored muscles were superficial and large, and therefore easy to locate with skin palpation. The choice of the muscles to be monitored was made by selecting the ones monitored in most works about hand muscular synergies. The results obtained in this work were aligned to previous works [16,18], thus suggesting the applicability of sEMG technique in extracting muscle synergies. Invasive EMG would have allowed to record muscle activity in a more precise way, but it was not possible to exploit this technique in this work for its invasiveness and for the type of task to be performed (i.e., grasping task requiring the interaction with objects that could hit the needles and generate discomfort and pain to the subjects).

Despite other more performing solutions, such as force sensors able to provide the pressure map, could have been used to record grasping forces, piezoresistive force sensors were selected to monitor the force exerted by the hand fingers and palm as these sensors respect the technical specifications to record hand grasping force and guarantee minimum encumbrance on the objects. In fact, this type of sensors has limited thickness, force sensitivity range of 0.2–20 N (i.e., the typical force range in activities of daily living [49,50]) and continuous force resolution. Each sensor was characterized using the Instron testing machine. Sensors were positioned on the objects according to the most frequent contact points between the hand and the objects [28] and by direct observation. This configuration allowed recording the force exerted by each finger and palm. Each subject was able to activate the sensors positioned on the objects during the tasks, and this demonstrates the correct sensors positioning. A previous work in literature exploited resistive sensors embedded in a glove to extract hand force synergies [15] and obtained significant and interesting results on force synergies. The choice of using force sensors directly placed on the objects, and not a glove like in [15], was driven by the fact that the glove setup would have been incompatible with the markers positioning to record hand kinematics and would hinder joint motion and negatively affect the grasp stability by reducing sensory feedback when grasping an object.

5. Conclusion

This work provided a comprehensive analysis of task-specific kinematic, force and muscle synergies and characterized generalized synergies in these domains that were shared between healthy volunteers. The types of grasps involved in the experimental protocol were classified according to the Cutkosky taxonomy [23], and power and precision grasps were selected to extract task-specific synergies. Both “pure” kinematic, force and muscle synergies and synergies extracted from combined datasets were analyzed, to investigate possible relations between variables involved in the grasp action in a multidimensional domain. Average generalized synergies were extracted after computing the CS index, which indicated that the first-order synergies were highly similar between subjects, thus suggesting the possibility of adopting generalized synergies to describe human grasping. What emerged was also that kinematic and force synergies were distinctive for the considered grasp classes. Synergies extracted from combined force and kinematic data showed that forces progressively gained importance in higher order synergies, and this suggests that the regulation of hand posture may not be independent from the regulation of contact forces. Simulations were conducted in Simulink and Simscape Multibody to generate power and precision grasps by activating pure kinematic synergies and combined kinematic and force synergies. Results showed that grasp stability significantly increased when grasps were generated through combined kinematic and force synergies, thus suggesting that the hand posture may be somehow modulated by the optimal distribution of contact forces to perform stable grasps.

Funding

This work was supported by the Italian Institute for Labour Accidents (INAIL) with 3D-AID and WI-FI MYOHAND projects (CUP: E59E19001460005) and by the European Union’s Horizon 2020 research and innovation programme under grant agreement No 899822 (SOMA project).

CRediT authorship contribution statement

Martina Lapresa: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Virginia Corradini:** Investigation, Writing – original draft. **Antonio Iacca:** Investigation, Writing – original draft. **Francesco Scotto di Luzio:** Conceptualization, Methodology. **Loredana Zollo:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Francesca Cordella:** Conceptualization, Writing – review & editing, Writing – original draft, Supervision, Project administration, Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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